The support–spectrum disjointness condition is not necessary

A literature-implied negative answer to arXiv:2310.03604

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Status: literature already implies a negative answer. No new theorem is claimed. The explicit example in arXiv:2509.04907 has a bounded embedding $K_u \hookrightarrow \mathcal{D}_\mu$ while $\operatorname{supp}\mu \cap \sigma(u) = \{1\} \neq \varnothing$.

1 The source problem

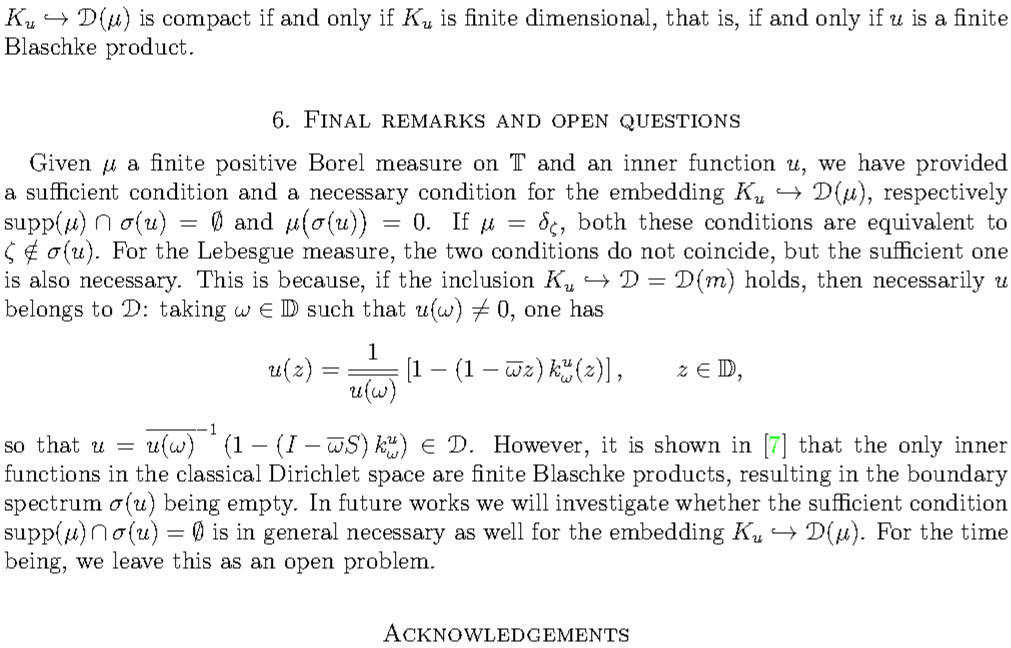
For an inner function u, write $K_u = H^2 \ominus uH^2$ and let $\sigma(u)$ be its boundary spectrum. For a finite positive Borel measure μ on $\mathbb{T}$, the harmonically weighted Dirichlet seminorm disintegrates as

$$D_\mu(f) = \int_{\mathbb{T}} D_\zeta(f)\, d\mu(\zeta).$$

Bellavita–Dellepiane prove that

$$\operatorname{supp}\mu \cap \sigma(u) = \varnothing \quad \Longrightarrow \quad K_u \hookrightarrow \mathcal{D}_\mu,$$

and that the embedding forces $\mu(\sigma(u)) = 0$. They ask whether the stronger support-disjointness condition is necessary in general.



$K_u \hookrightarrow \mathcal{D}(\mu)$ is compact if and only if K_u is finite dimensional, that is, if and only if u is a finite Blaschke product.

6. Final remarks and open questions

Given μ a finite positive Borel measure on $\mathbb{T}$ and an inner function u, we have provided a sufficient condition and a necessary condition for the embedding $K_u \hookrightarrow \mathcal{D}(\mu)$, respectively $\operatorname{supp}(\mu) \cap \sigma(u) = \emptyset$ and $\mu(\sigma(u)) = 0$. If $\mu = \delta_\zeta$, both these conditions are equivalent to $\zeta \notin \sigma(u)$. For the Lebesgue measure, the two conditions do not coincide, but the sufficient one is also necessary. This is because, if the inclusion $K_u \hookrightarrow \mathcal{D} = \mathcal{D}(m)$ holds, then necessarily u belongs to $\mathcal{D}$: taking $\omega \in \mathbb{D}$ such that $u(\omega) \neq 0$, one has

$$u(z) = \frac{1}{\overline{u(\omega)}}\left[1 - (1 - \overline{\omega} z)\, k^u_\omega(z)\right], \qquad z \in \mathbb{D},$$

so that $u = \overline{u(\omega)}^{-1}\left(1 - (I - \overline{\omega} S)\, k^u_\omega\right) \in \mathcal{D}$. However, it is shown in [7] that the only inner functions in the classical Dirichlet space are finite Blaschke products, resulting in the boundary spectrum $\sigma(u)$ being empty. In future works we will investigate whether the sufficient condition $\operatorname{supp}(\mu) \cap \sigma(u) = \emptyset$ is in general necessary as well for the embedding $K_u \hookrightarrow \mathcal{D}(\mu)$. For the time being, we leave this as an open problem.

Acknowledgements

Figure 1: The exact open problem on source PDF page 15 of arXiv:2310.03604v3.

2 The later explicit example

Bellavita–Dellepiane–Hartmann–Mashreghi, arXiv:2509.04907, take the singular inner function

$$u(z) = \exp\left(\frac{z+1}{z-1}\right), \qquad \sigma(u) = \rho(u) = \{1\}.$$

Its Clark atoms are

$$\zeta_n = \frac{2\pi i n + 1}{2\pi i n - 1}, \qquad n \in \mathbb{Z},$$

and they define

$$\mu = \sum_{n\in\mathbb{Z}} \frac{1}{|u'(\zeta_n)|^2}\,\delta_{\zeta_n}.$$

The paper verifies the required potential estimate and, in Theorem 8.3, proves

$$\mathcal{H}\left(\frac{1+u}{2}\right) = \mathcal{D}_\mu.$$

$$|1-u(z)|^2 V_\mu(z) = \lim_{N\to+\infty} \Phi_N(z) \le 4M.$$

□

8.1. **An explicit example of a one-component inner function satisfying** (2.6)**.** We prove that the singular inner function u associated to δ_1,

$$u(z) = \exp\left(\frac{z+1}{z-1}\right), \qquad z \in \mathbb{D},$$

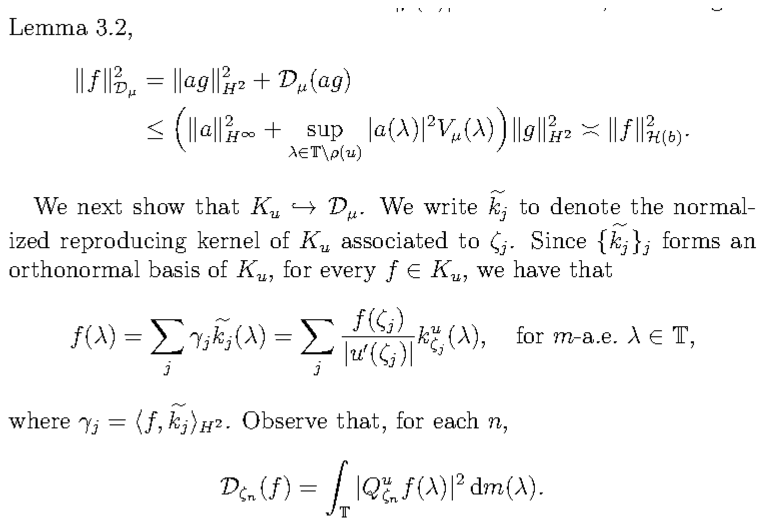
and the measure

$$\mu = \sum_n \frac{1}{|u'(\zeta_n)|^2}\delta_{\zeta_n},$$

where ζ_n are the Clark atoms of u, satisfy the potential condition $\sup_{\mathbb{D}} |1-u|^2 V_\mu < \infty$. This provides an explicit example of the equality $\mathcal{H}(b) = \mathcal{D}_\mu$, where μ is a measure with infinitely many atoms.

Figure 2: The construction on supporting PDF page 35 of arXiv:2509.04907v1.

The implication needed here is explicit in that paper. In the proof of its Theorem 6.3, after treating the aH^2 component, the authors separately prove $K_u \hookrightarrow \mathcal{D}_\mu$.


Lemma 3.2,

$$\|f\|^2_{\mathcal{D}_\mu} = \|ag\|^2_{H^2} + \mathcal{D}_\mu(ag)$$
$$\leq \Big(\|a\|^2_{H^\infty} + \sup_{\lambda\in\mathbb{T}\setminus\rho(u)} |a(\lambda)|^2 V_\mu(\lambda)\Big)\|g\|^2_{H^2} \asymp \|f\|^2_{\mathcal{H}(b)}.$$

We next show that $K_u \hookrightarrow \mathcal{D}_\mu$. We write $\widetilde{k}_j$ to denote the normalized reproducing kernel of K_u associated to ζ_j. Since $\{\widetilde{k}_j\}_j$ forms an orthonormal basis of K_u, for every $f \in K_u$, we have that

$$f(\lambda) = \sum_j \gamma_j \widetilde{k}_j(\lambda) = \sum_j \frac{f(\zeta_j)}{|u'(\zeta_j)|} k^u_{\zeta_j}(\lambda), \quad \text{for } m\text{-a.e. } \lambda \in \mathbb{T},$$

where $\gamma_j = \langle f, \widetilde{k}_j\rangle_{H^2}$. Observe that, for each n,

$$\mathcal{D}_{\zeta_n}(f) = \int_{\mathbb{T}} |Q^u_{\zeta_n} f(\lambda)|^2 \, \mathrm{d}m(\lambda).$$

Figure 3: The component-embedding step on supporting PDF page 27.

3 Why this negates the source condition

Every coefficient of μ is positive. Moreover,

$$\zeta_n - 1 = \frac{2}{2\pi i n - 1} \longrightarrow 0 \quad (|n| \to \infty),$$

so every neighbourhood of 1 has positive μ-mass. Hence $1 \in \operatorname{supp}\mu$. No atom is placed at 1, and the paper computes

$$\frac{1}{|u'(\zeta_n)|} = \frac{2}{4\pi^2 n^2 + 1},$$

so μ is finite and $\mu(\{1\}) = 0$. Therefore

$$K_u \hookrightarrow \mathcal{D}_\mu, \qquad \mu(\sigma(u)) = 0, \qquad \operatorname{supp}\mu \cap \sigma(u) = \{1\}.$$

This is exactly a counterexample to necessity of the source's sufficient condition, while sharply respecting its proved necessary condition.

4 Crosswalk and conclusion

The notation $\rho(u)$ in arXiv:2509.04907 is the same closed boundary spectrum that arXiv:2310.03604 denotes by $\sigma(u)$: for inner u it is the set of boundary points where $\liminf_{z\to\zeta} |u(z)| = 0$. Thus there is no hypothesis mismatch. The later article does not advertise its example as an answer to the 2023 final question, so this packet classifies the answer as `literature_implied`, not as a new counterexample.

References

[1] C. Bellavita and E. Dellepiane, *Embedding model and de Branges–Rovnyak spaces in Dirichlet spaces*, Complex Variables and Elliptic Equations 70 (2025), 228–246; arXiv:2310.03604.

[2] C. Bellavita, E. Dellepiane, A. Hartmann, and J. Mashreghi, *Infinitely supported harmonically weighted Dirichlet spaces which are de Branges–Rovnyak spaces*, arXiv:2509.04907 (2025).